Beyond Triage: Coordinating an Unreliable Language Model into a Dependable Message-Based Health Triage Pipeline

with active follow-up as the mechanism that turns uncertainty into a question

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Abstract

Small language models are cheap to run and understand colloquial, code-mixed symptom descriptions well, which makes them attractive for health triage over SMS or voice in settings with few clinicians. They are also unreliable decision-makers: asked directly to sort 304 held-out cases into four urgency levels, a 3-billion-parameter model labelled 251 of them "severe", missed 16.9% of the cases that were urgent, and over-called 82.7% of the ones that were not. This paper describes a software framework that keeps the model where it is useful, reading free text, and surrounds it with deterministic structure that makes the overall system dependable. The framework coordinates four stages: structured extraction from free text, a classifier with a calibrated abstain decision, urgency prioritisation, and resource-aware routing. Its central mechanism is active follow-up: when the classifier cannot commit, the system asks one targeted yes/no question and decides again. Evaluated out-of-sample on a 41-condition demonstration dataset, the full pipeline reduces missed urgent cases to zero (a bound that holds by construction and is verified across a sweep of its one free parameter), and the follow-up loop then recovers most of the over-caution that bound costs, cutting over-calls from 21.6% to 5.6% in a mean of 2.15 questions. Two findings came from measuring rather than reasoning. First, an information-gain question selector lost to random selection until the classifier was taught to use a "no"; after that it resolved 74.8% of uncertain cases, 3.9 times its previous rate. Second, what looked like a 13.3% extraction failure rate of the language model was a single schema bug in the code around it. We make no clinical claim: urgency labels were assigned by the authors, not clinicians, and every rate reported is a property of the framework's uncertainty handling relative to those labels. Code, data provenance, and every evaluation harness are public, and the deterministic harnesses reproduce byte for byte.

Keywords: health triage, selective prediction, conformal prediction, active follow-up, language models, low-resource settings, software architecture, SDG-3.

1 Introduction

A community health worker in a village receives a message: "bachche ko fever aur loose motion hai, 2 din se". Somebody has to decide whether this child should be seen today, taken to the district

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hospital, or kept at home with fluids. In much of rural India that decision is made by someone with a few weeks of training, a phone, and no clinician within reach. The question this project started with was whether a small language model running on ordinary hardware could help with that decision.

The answer we arrived at is that the model should not make the decision at all. It should read the message. Everything after that should be deterministic, inspectable, and built so that uncertainty produces a question rather than a guess.

That position was not where we began. The first version of the system prompted a model with the symptoms and the four urgency tiers and asked it to pick one. Section 5.1 reports what happened when we measured that properly: the model collapsed onto a single answer, calling 251 of 304 cases "severe", and was at the same time unsafe (it missed 16.9% of urgent cases) and uninformative (it over-called 82.7% of routine ones). Writing "when uncertain, choose the more severe option" into the prompt did not change this. The structure around the model is what changed it.

This paper describes that structure. It is a computer-science contribution, not a medical one, and it is a contribution about coordination rather than about any single component. The classifier is a Naive Bayes model. The abstain gate is split conformal prediction. The prioritiser is an AHP score. None of these is new. What we think is worth writing down is how they are joined so that no one of them is responsible for safety alone, and in particular the loop that converts "the system is unsure" into "ask this one question, then decide again".

Contributions.

1. A framework that coordinates free-text extraction, a safety-calibrated abstain decision, urgency prioritisation, and resource-aware routing into one pipeline for message-based triage (Section 3), with the shared typed case record as the contract every module speaks.

2. Active follow-up as the mechanism linking uncertainty to action, evaluated out-of-sample against two baselines and under two evidence models (Section 5.4). The result that the question selector loses to random selection until denials are used as evidence is, we think, the most useful thing in the paper.

3. An evaluation that reports the safety property as what it is, a distribution-free bound entailed by worst-case labelling over a conformal set, and that measures its premise (coverage) and its price (over-calls) rather than presenting the zero as a discovery (Section 5.3).

4. A demonstration that the design is extensible: an outbreak-surveillance module was added with zero edits to the decision path, enforced by a test rather than a convention (Section 5.8).

5. Several negative and corrective results, reported as such: a tuning constant that does nothing, a conformal fallback that fires on a third of cases at the conventional α, an absence-evidence implementation that made the system more alarmed the more a caller denied, and a model "failure rate" that was a bug in our own code.

Everything is reproducible from the repository. The deterministic evaluation harnesses produce identical output on every run; the three that call the language model are reproducible up to the model's own determinism, which we measured at 1.000 at temperature zero.

2 The problem, stated as four computational tasks

We treat message-based triage as four problems that have to be solved together, with one mechanism linking the second to the rest.

1. **Extraction under language variation.** The input is a short message in English, Hindi, or a mixture, written by a layperson. "Pet dard aur ulti" and "stomach ache with vomiting" must become the same structured record.
2. **Classification with a calibrated abstain.** Given the record, decide an urgency tier, or decline to decide. The system must know when it does not know, and the rate at which it is confidently wrong has to be controlled, not hoped for.
3. **Urgency prioritisation.** Among cases at the same tier, which should be attended first.
4. **Resource-aware routing.** Which facility can actually treat this case, is open, and is reachable.

Active follow-up links problem 2 to the others. When the classifier abstains, the system does not escalate blindly and does not guess. It selects the single yes/no question whose answer would most reduce its uncertainty, asks it, folds the answer back into the record, and classifies again, up to three times.

What dependable means here. Two rates are used throughout the paper. Both are engineering properties of how the pipeline handles its own uncertainty, measured against the labels of a demonstration dataset (Section 4.1); neither is a clinical judgement.

- **MISS** (misses urgent): a case whose dataset label is urgent (EMERGENCY or SEVERE) was surfaced at a lower tier. $n = 142$.
- **OVER** (over-calls): a case whose dataset label is routine (MODERATE or MILD) was surfaced at a higher tier. $n = 162$.

The clinical literature calls these under-triage and over-triage. We use the plainer terms because we want to be clear that what is being measured is the framework's consistency with a set of labels, and not patient outcomes.

Design goals. Four properties guided every decision. The language model is called as few times as possible and never for the decision itself. Every non-model stage is deterministic and auditable. Uncertainty is surfaced as a question, not hidden in a confident label. And the parts are separated by typed interfaces so that one can be replaced or a new one added without rewriting the core.

3 The framework

Figure 1 shows the pipeline. A message enters at the top; a per-role view (caller, health worker, doctor, auditor) leaves at the bottom. The important arrows are the one that returns from the follow-up loop to the classifier, and the escalation rules that cut across every branch. No single module decides safety alone.

3.1 Extraction (Agent 1)

The only language-model call on the routine path. A local `llama3.2:3b` served by Ollama reads the message and returns a structured record whose clinical fields are explicitly three-valued: `True` (reported present), `False` (reported absent), `None` (not mentioned). The distinction between "absent" and "not mentioned" is the most load-bearing rule in the codebase. Nothing downstream is allowed to collapse `None` into `False`, because doing so turns "we do not know" into "we checked and it is fine".

Three backends sit behind one interface: the local model, a hosted model, and a regex stand-in for channels where inference is not viable. The record is validated at the boundary and repaired where the model's output is recoverable. One such repair turned out to matter a great deal (Section 5.1).

Symptoms come back as a list of free-text phrases. A code-mixed clause splitter separates conjunctions in either script (*aur* in either script, *and*) before matching. Each phrase is then mapped to one of 131 canonical symptom tokens by nearest-neighbour search over multilingual sentence embeddings (`paraphrase-multilingual-MiniLM-L12-v2`, FAISS index). The confidence threshold for that match was calibrated on an 80-item labelled sample (97.5% accuracy at the shipped value) and a test asserts the shipped constant equals the calibrated one, so the two cannot drift apart silently.

3.2 The decision gate

Adult cases go to a Naive Bayes classifier over the 41 conditions in the demonstration dataset. We did not want its argmax. We wanted a set of plausible conditions with a guarantee on how often the truth is in it. Split conformal prediction gives that. The classifier is fitted on four of five profile folds; the held-out fold provides nonconformity scores $(1 - \hat{P}(\text{true class}))$, and the $(1-\alpha)$ quantile $\hat{q}$ of those scores becomes the inclusion threshold at inference: a condition is in the prediction set $S(x)$ if its posterior exceeds $1-\hat{q}$. The construction is the least-ambiguous-set classifier of Sadinle et al. [1]. Marginal coverage $P(y \in S(x)) \geq 1 - \alpha$ holds without distributional assumptions, provided calibration and test points are exchangeable. The nonconformity scores are computed out-of-fold, by each fold's own model, so the calibration is as leak-free as the rest of the cross-validation.

The size of S is the decision:

- $|S| = 1$: CONFIDENT. The condition's severity tier is the label.
- $|S| \in \{2, 3\}$: UNCERTAIN. The label is the most severe tier among the conditions in S, and the follow-up loop is entered.
- $|S| \geq 4$: ABSTAIN. If any condition in S is urgent, the label is again the worst case; otherwise the case is marked for a human. The loop is entered.

Worst-case labelling over the set is the mechanism that carries the safety result, and Section 5.3 says exactly what it does and does not guarantee.

Two things we disclose now because they shaped the results. First, when the LAC threshold admits no condition at all, the implementation returns every surviving candidate rather than an empty set; at the conventional $\alpha = 0.10$ this fallback fires on 33.9% of cases. Second, the classifier originally scored only present symptoms. A denied symptom contributed nothing. Section 5.4 is largely about what that cost and what fixing it required.

Paediatric cases (2 to 59 months) do not go through the classifier. They go through a small hand-coded slice of the WHO IMNCI danger-sign logic [7]: convulsions, inability to drink, lethargy, and the like route directly to EMERGENCY. A danger-sign override is applied across every adult branch as well. This is deliberately a small slice, not an IMNCI implementation, and we say so in Section 7.

3.3 Active follow-up

On the UNCERTAIN and ABSTAIN branches the system has a set of plausible conditions and does not want to escalate blindly. It asks a question instead.

The question is selected, not generated. A selector scores every symptom token not yet known for the case by how much either answer would shrink the prediction set, using the classifier's actual posteriors over S and the dataset's symptom frequencies per condition. Tokens a caller cannot answer about (coma, acute liver failure, receiving blood transfusion, twelve in all) are excluded from the candidate pool. If an urgent condition is in S, its characteristic discriminators are preferred. The language model only renders the chosen token into a sentence in the caller's language; it cannot change which symptom is asked about. This is what makes the selection policy evaluable deterministically.

The reply is parsed as yes, no, or not understood. Hedges ("maybe", "pata nahi") are deliberately not coerced; the caller is re-asked. A "yes" adds the token to the record. A "no" adds it to a separate list of absent tokens, which the classifier now uses as evidence, with a restriction that Section 5.4 explains. The classifier runs again on the updated record, and the loop continues until the set is a singleton, the question budget (three) is spent, or no discriminating question remains.

The loop is stateless by construction: everything it needs is in the record and the question/answer trail, so the same code runs as a blocking loop in a test harness and one message at a time over SMS.

3.4 Prioritisation and routing

Within a tier, an Analytic Hierarchy Process score orders cases for attention. Six attributes (complication risk, time-to-treatment sensitivity, severity, age, onset, transmissibility) are weighted by the principal eigenvector of a pairwise comparison matrix [6]. We make one small point about this: the weights, $\lambda_{\max}$ and the consistency ratio are derived from the matrix at import time by power iteration, not transcribed into the code. An earlier version quoted a consistency ratio of 0.0205 that could not be reproduced from the matrix; the derived value is 0.0198, and a test refuses to import the module if the ratio exceeds 0.10. The AHP score is a within-tier priority, not a triage label. It never changes the tier.

Routing filters facilities by what the case needs (an EMERGENCY needs a facility that can handle one; an urgent case needs one that is open), then scores the survivors by haversine distance from the caller's village. The provider interface has an offline mock default and an OpenRouteService adapter. A handoff report for the receiving clinician is generated by the model, with the dataset diagnosis suppressed whenever it was not the basis of the decision (danger-sign and abstain paths), because an earlier version produced "cervical spondylosis, apply heating pad" for a convulsing infant.

3.5 Extension seams

A framework rather than an application has to be something another person can adopt a piece of, swap a piece of, or grow. Four seams are exposed on purpose. The model backend is an interface with three implementations. The translation and routing providers are interfaces with no-op or offline defaults. And every module reads and writes one typed case record, so a new module consumes the same stream the existing ones produce. Section 5.8 reports what it took to add a second module through those seams.

Where the language model is and is not. Counting calls on the shipped path: one for extraction on every case; up to three more for phrasing follow-up questions, only when the gate abstains; one for the handoff report, only on urgent cases; and one for a surveillance summary, only when an alert fires. The decision, the prioritisation, the routing, the reply parsing, and the question selection are all deterministic. Section 5.6 gives the latencies; the framework's own logic is sub-second beside a single model call of about four seconds.

4 Evaluation setup

4.1 The demonstration dataset, and what it is not

The knowledge base is a public, Kaggle-style tabulation of 41 conditions against symptoms: 4,920 rows, 120 per condition, 3 to 17 symptoms per row, 131 distinct symptom tokens after normalisation. It is a synthetic teaching dataset. It was not collected from patients and does not represent real presentations. We use it as a sandbox in which the framework's behaviour can be isolated from real-world confounders, and we call it a demonstration dataset throughout.

Two properties of it matter for reading the results.

The 120 rows per condition collapse to only 304 unique (condition, symptom-set) profiles, between 5 and 10 per condition. Every split in this paper is therefore on profiles, never on rows; a row-level split would put near-duplicates on both sides and leak.

The symptom–condition associations are effectively binary. Across all 321 (condition, token) pairs, 85.4% of associations hold in at least 95% of that condition's rows, 100% hold in at least 70%, and none fall below. Real clinical associations are probabilistic. This makes the discrimination problem easier than reality, and it means the follow-up results in particular should not be extrapolated to real data.

The urgency labels. The source dataset has no severity column. The four-tier labels (EMERGENCY 2, SEVERE 16, MODERATE 18, MILD 5) were assigned by the authors, mapping each condition name onto the tiers the system outputs. No clinician reviewed them. They are not derived from ESI, IMNCI, or any published standard. Every safety number in this paper is measured relative to these labels, so it is a statement about the framework's internal consistency and uncertainty handling, and not evidence of clinical safety. We regard this as the single most important limitation of the work and repeat it in Section 7. One structural consequence: only two EMERGENCY conditions exist, so the urgent population ($n = 142$) is dominated by SEVERE.

4.2 Protocol

Leave-profiles-out five-fold cross-validation over the 304 profiles. Each held-out profile is cut to a random subset of 2 to 4 symptoms before being presented to the system, because a person describing a problem mentions a few symptoms, not their complete record. The full profile is retained as ground truth. All harnesses use seed 42.

Where a harness needs the language model, it is the same local `llama3.2:3b` at temperature zero. Three harnesses call the model; the rest are fully deterministic (no model, no embedding index) and reproduce byte for byte. The suite that locks every mechanism described in Section 3 has 627 passing tests.

4.3 Three comparators

- **LLM-only.** The model is given the same symptoms and a prompt that defines the four tiers and states the safety convention ("when uncertain, choose the more severe"). It is asked for one tier. This is the "why not just prompt it?" baseline and we tried to make it fair.
- **NB top-1.** The Naive Bayes argmax with none of the framework's machinery: no emergency boost, no conformal set, no worst-case labelling. This is an independent baseline, not an ablation. (An earlier version of the harness scored the baseline and an ablation with the same code and reported one number twice. A test now asserts the comparators are distinct computations.)
- **Ablations** of the full system with one mechanism removed at a time.

5 Results

5.1 Prompting the model directly does not work

Table 1 is the motivation for everything else.

Table 1: A single prompted model versus the framework on the same 304 held-out cases.

System	MISS ↓	OVER ↓	Exact tier
LLM-only (`llama3.2:3b`, safety prompt)	0.169 (24/142)	0.827	0.454
Framework, full pipeline	0.000	0.216	

Of 304 replies, 251 were SEVERE, 36 MODERATE, 11 MILD, 6 EMERGENCY. The model is simultaneously unsafe and uninformative because it is barely discriminating. Output format was never the issue: every one of 304 replies was a valid tier, with no transport errors, and label stability at temperature zero across 25 cases times three calls was 1.000. The problem is the decision, not the plumbing.

What the extraction "failure rate" turned out to be. Before the decision results, one finding about the extraction stage. On a labelled fixture of Hindi, Hinglish and English messages, the first measurement of Agent 1 gave 86.7% schema-valid output, 80.0% expected-token recall, and a 13.3% failure rate, and the natural reading was that a 3B model is unreliable at structured

extraction. It was not. The repair function for malformed severity strings guarded on `sev is not None`, so an explicit `"severity": null` passed through to a non-optional field and the whole case was discarded. The model emits exactly that for bare complaints ("chest pain", "headache", "stomach pain", "constipation"), and those four inputs were 100% of the failures. After the fix, and two prompt corrections found the same way ("mild fever" was being split into `fever` plus `severity:mild`, and one-word messages were dropped), the fixture scores 100% schema-valid, 100% recall, 0% failure. What looked like a capability limit of the model was an integration defect in the layer around it, and only measurement could tell the two apart.

5.2 The framework, one layer at a time

Table 2 adds one mechanism per row, scored on the same held-out cases.

Table 2: The journey from a prompted model to the shipped pipeline. Each row adds one layer to the row above.

Configuration		MISS ↓	OVER ↓
C0	LLM-only	0.1690	0.8272
C1	+ structured classifier (NB top-1)	0.0915	0.0679
C2	+ conformal abstain gate	0.0634	0.0679
C3	+ worst-case labelling over the set (full system)	0.0000	0.2160
C4	+ active follow-up (shipped)	0.0211	0.0556

Read top to bottom. Structure removes most of the naive model's failure (C0 to C1). The abstain gate removes more (C1 to C2). Worst-case labelling removes the rest, and pays for it with caution: over-calls rise from 6.8% to 21.6% (C2 to C3). Active follow-up then buys most of that caution back: over-calls fall to 5.6% while misses stay inside the designed bound (C3 to C4).

Table 3 asks which layer is load-bearing by removing one at a time.

Table 3: Component ablation. "Confident" is the fraction of cases on which the system committed to a single condition rather than asking or abstaining.

Configuration	MISS	OVER	Confident
NB top-1, no framework machinery	0.0915	0.0679	100%
+ conformal set, no worst-case labelling	0.0634	0.0679	66.1%
+ worst-case labelling (full system)	0.0000	0.2160	66.1%
full system, emergency-boost constant removed	0.0000	0.2160	65.8%

Two findings. Worst-case labelling is what carries the result: removing only that layer moves misses from 0.000 to 0.063. And a negative result: the classifier had a hand-set `EMERGENCY_SAFETY_FACTOR` of 3.0 that boosted the prior of the two emergency conditions, defended in a comment as necessary for recall. Removing it changes nothing. Swept from 1.0 to 10.0 with the conformal gate re-calibrated at each value, it never reduces misses and at higher values slightly increases over-calls (0.210 at 1.0, 0.216 at 3.0 and above). Worst-case labelling already does the job the constant was meant to do. It is not load-bearing and is a candidate for deletion; a test now locks that finding so it cannot be quietly reintroduced as a safety measure.

5.3 The safety property is a bound, not a measurement

This distinction decides whether the central claim survives review, so we state it rather than leave it for a reader to notice.

The framework labels a multi-condition prediction set with the most severe tier in that set. If the true condition is in the set, the surfaced tier is at least as severe as the truth, and a miss is impossible. A miss can occur only when the conformal set does not contain the truth, which split conformal prediction bounds. With a conformal set $S(x)$ of marginal coverage at least $1 - \alpha$, and worst-case-tier labelling over $S(x)$:

$$P(\text{miss}) \;\le\; P\big(y \notin S(x)\big) \;\le\; \alpha.$$

So "0 of 142" in Table 2 is entailed by the measured coverage of 1.000; it is not independent evidence. Presented correctly this is stronger than an empirical zero: it is a distribution-free guarantee that holds by construction and is tunable through α. What the harness measures empirically is the premise (coverage) and the price (over-calls).

Table 4 sweeps α and checks the bound at every level.

Table 4: Sensitivity to the one free parameter. The bound $P(\text{miss}) \le \alpha$ holds in every row. "Fallback rate" is the fraction of cases on which the LAC threshold admitted no condition and the implementation returned every surviving candidate instead.

α	threshold $1 - \hat{q}$	coverage	MISS	OVER	confident	fallback rate
0.01	0.9682	0.984	0.000	0.173	0.691	0.000
0.02	0.8940	0.961	0.007	0.142	0.743	0.000
0.05	0.5424	0.957	0.028	0.124	0.813	0.135
0.10 (shipped)	0.0216	1.000	0.000	0.216	0.661	0.339
0.15	0.0102	1.000	0.000	0.235	0.612	0.388
0.20 / 0.30	0.0000	1.000	0.000	0.265	0.582	0.418

Two things this table discloses that a reviewer would otherwise find for us.

The fallback has to be reported. At $\alpha = 0.10$ the LAC threshold is 0.0216, and on 33.9% of cases no condition clears it, so the gate returns everything. Those decisions come from the fallback plus worst-case labelling, not from conformal filtering proper. The coverage of 1.000 at the shipped setting is partly an artefact of that. Describing the behaviour at $\alpha = 0.10$ as a conformal gate that abstains would overstate it; it is closer to a binary gate (confident singleton, or "return everything").

The shipped α is dominated. $\alpha = 0.01$ is at least as good on every axis and better on two: the same zero misses, lower over-calls (0.173 against 0.216), a higher confident rate (0.691 against 0.661), and no fallback at all. Nominal coverage is slightly under-attained at the tightest settings (0.984 against 0.99; 0.961 against 0.98) because coverage here is marginal and finite-sample, with 304 calibration points. We publish both settings so a reader can choose. $\alpha = 0.10$ is retained in the shipped configuration as the conventional value, and because the point of the sweep is that the trade-off is designed rather than tuned.

5.4 Active follow-up

This is the mechanism the paper is about, and until recently it had no quantitative evaluation. It is now measured out-of-sample on the same held-out cases with a three-question budget.

How answers are obtained without inventing data. Each held-out case is a real profile from the dataset with a complete, recorded symptom list. The system is shown only the 2 to 4 symptom subset. When it asks "does the patient have X?", the answer is read off that same profile: yes if X is in the full record, no otherwise. Nothing is generated or simulated.

What is being evaluated. The question is chosen by the deterministic selector, not by the model; the model only phrases it. So the selection policy can be compared against two baselines, most-frequent (ask about the token most common in the prediction set's conditions) and random (any askable, unknown token), holding everything else fixed. We also compare two evidence models in the classifier: ignores "no", the original behaviour in which only present tokens were scored, and counts "no", the shipped behaviour.

Table 5 is a two-part finding.

Table 5: The follow-up loop: question-selection policy against evidence model. "Resolved" is the fraction of uncertain cases ending with a singleton set; "resolved correctly" also requires the singleton to be the true condition.

Evidence model	Selector	Resolved	Mean Q	$\|S\|$ before $\rightarrow$ after	Resolved correctly	MISS	OVER
ignores "no"	information-gain	0.194	2.71	$4.33 \rightarrow 3.30$	0.194	0.000	0.216
ignores "no"	most-frequent	0.097	2.97	$4.33 \rightarrow 3.37$	0.058	0.007	0.191
ignores "no"	random	0.291	2.68	$4.33 \rightarrow 3.43$	0.272	0.000	0.142
counts "no" (shipped)	information-gain	0.748	2.15	$4.33 \rightarrow 1.67$	0.670	0.021	0.056
counts "no"	most-frequent	0.427	2.62	$4.33 \rightarrow 1.97$	0.340	0.007	0.105
counts "no"	random	0.583	2.45	$4.33 \rightarrow 1.94$	0.544	0.007	0.086

First, the clever selector was worse than random. Under the original evidence model, information-gain selection resolved 19.4% of uncertain cases and random selection resolved 29.1%. The cause is a design inconsistency rather than a bug in either part. The selector picks the question whose answer, yes or no, would most shrink the set; its objective models P(absent) explicitly. But the classifier scored present tokens only, so a "no" taught it nothing. Half of the information each question was selected to obtain was being discarded, and the optimisation was wasted. Random selection did not suffer from this because it was not optimising for anything.

Second, make "no" informative and the selector dominates. A Laplace-smoothed Bernoulli absence term in the classifier raises resolution from 0.194 to 0.748, a factor of 3.9; correct resolution from 0.194 to 0.670; and the mean number of questions falls from 2.71 to 2.15. The selector now beats random (0.583) and most-frequent (0.427) by clear margins. The net effect on triage is a favourable trade: over-calls fall from 21.6% to 5.6%, because resolved cases no longer need worst-case escalation, at a cost of 2.1% misses, still inside the α bound, with the true condition retained in the prediction set 97.4% of the time.

A pathology found while shipping the fix. The obvious implementation applies the absence penalty across all 41 conditions. That is correct Bayes for a true generative model, but this classifier scores presence only, so the two terms are not on the same footing: after a few denials, the penalties on the plausible conditions outweigh the thin positive signal, the posterior flattens, and conditions the denied symptoms have nothing to do with rise simply because they were penalised least. Measured directly: a case with `[cough]` and `chest_pain`, `phlegm`, `high_fever` denied produced a 28-condition prediction set containing paralysis and positional vertigo, which worst-case labelling then escalated to EMERGENCY. Denials made the system less certain and more

alarmed, the exact opposite of their purpose.

The fix restricts the penalty to the candidate pool that positive evidence already supports: it conditions on the differential the patient's actual symptoms imply. A "no" can then narrow or re-rank the set but can never introduce an unsupported condition and can never raise the tier. Those two properties are asserted by tests. The restricted version also scored better than the unrestricted one (resolution 0.718 to 0.748, final set size 1.97 to 1.67). We think it is worth stating plainly that the unrestricted version would have passed a casual review, because its aggregate numbers looked fine. A monotonicity check on one hand-built case is what exposed it.

Question quality. Selection was originally driven by discriminative power alone, with 32 phrasing templates over the 131-token vocabulary. The result was that 53.6% of questions fell back to a generic "Is the patient experiencing {token}?" and the loop asked laypeople things like "Is the patient experiencing acute liver failure?" and "…coma?": statistically sound, clinically incoherent, and unfixable by rewording because the chosen token is not something a caller can report. Twelve diagnosis, history and instrumented tokens are now excluded from the candidate pool, and templates cover all 119 askable tokens. Measured afterwards: generic fallback 0%, unanswerable tokens selected 0.

5.5 End to end: real text in, decision out

Every harness above scores symptom tokens, which excludes the framework's first two stages. This one closes the gap. The same folds and the same labelling logic are run twice per case: ORACLE on the ground-truth tokens, and END-TO-END on the tokens the pipeline actually recovers from rendered text, so the difference is attributable to extraction alone.

Table 6: The composed pipeline from free text to tier. In the lay condition every symptom is replaced by a real caregiver phrasing, rotated across English, Hindi (Devanagari) and Hinglish, so each message is code-mixed.

Rendering	n	token recall	exact match	MISS	agreement w/ oracle	extraction cost (OVER)
canonical English (ceiling)	304	0.993	0.980	0.000	0.997	−0.006
lay, code-mixed	83	0.962	0.892	0.000	0.976	+0.053

The headline is that misses are zero end to end in both conditions: the bound survives real extraction through code-mixed lay language. What extraction costs is precision, 5.3 points of over-calls and 2.4% label disagreement with the oracle.

One caveat belongs next to the table. The lay subsample is selected by which symptoms have labelled phrasings, not at random, so its absolute over-call rate (0.842) is dominated by whichever conditions happen to have them; the oracle on those same 83 rows is already at 0.789. Compare the delta, not the absolute. The harness writes this rule into its own output.

5.6 Robustness, reliability, cost

Repeat agreement and paraphrase agreement are reported separately on purpose. The first isolates model nondeterminism; the second is the language-robustness claim. Collapsing them would let a perfect repeat score stand in for a language result, which is what an earlier "token-order consistency = 1.0" metric did before we recognised it as tautological on a sorted bag-of-tokens model and

Table 7: Robustness to how the message is written, and reliability of the model-facing stages.

Property	Result
Same message submitted three times, same decision	1.000
Three different real phrasings of the same content, same decision	0.750
Different 2–4 symptom subset of the same profile, same tier (277 profiles)	0.866
Replies not understood (37 definite forms, incl. Hindi and Hinglish)	0.000
Replies read as the opposite answer	0.000
Hedges ("maybe", "pata nahi"; 11 forms) coerced into an answer	0.000
Valid output format, LLM-only baseline	100% (0/304)
Label determinism at temperature zero	1.000
Agent 1 schema-valid output (after fix)	1.000 (was 0.867)
Agent 1 expected-token recall (after fix)	1.000 (was 0.800)

removed it. Subset robustness at 0.866 is reported precisely because it is imperfect and therefore informative.

Refusing to interpret a hedge is a safety property, not a nicety. "Maybe" must stay unparseable so the caller is re-prompted rather than have an invented clinical fact written into their record. The reply list is authored, so this measures parser coverage over documented reply forms, not observed caller behaviour.

Cost. On a typical routine case the pipeline makes one model call (extraction). On a typical urgent case, two (plus the handoff summary). Follow-up phrasing adds zero to three calls, only when the gate abstains; the surveillance summary is event-driven and fires only on an alert. The deterministic stages take about 55 ms (decision), 0.06 ms (routing) and 0.23 ms (surveillance pass) against roughly 3,800 ms for the single extraction call. Latencies are machine-dependent and should be compared across stages, not quoted as absolutes; the point is that everything the framework adds is either sub-second or conditional on uncertainty.

5.7 Prioritisation and routing

These are the least novel parts of the system and are reported for completeness, because an earlier draft claimed four computational contributions while evidencing two.

The discrimination figure is the one that matters for AHP. Its declared role is ordering patients within a tier, so a scorer that tied everyone would be a relabelling of the tier it sits on. At 0.659 it strictly orders two-thirds of same-tier pairs. No clinical-correctness claim is made for the ordering; the dataset has no ground-truth dispatch order to compare against.

5.8 Extensibility: a second module with zero core edits

The claim that this is a framework and not one application has to be evidenced structurally. We added an outbreak-surveillance module as the test. It watches the stream of recorded cases for unusual clusters in an area and raises an alert with a recommended action.

The module plugs in at a defined seam, the case stream, and reuses the shared case contract. It calls the model only when an alert fires, never per case, and falls back to a deterministic summary when the model is unavailable. That a second capability was added without touching the decision

Table 8: Prioritisation and routing. The facility network is a 10-facility synthetic demonstration set with a mock distance provider; this evaluates the logic, not real-world travel.

AHP (1,025 cases)	
Within-tier pair discrimination	0.659
Distinct scores per tier	3–5
Monotone in the severity attribute	yes
Danger sign always forces top priority	yes
Consistency ratio (derived at import)	0.0198
Routing (10 scenarios, 5 villages × 2 urgency levels)	
Optimality versus exhaustive brute force	1.000
EMERGENCY eligibility respected	1.000
urgent open-hours respected	1.000
Mean / max ETA	12.0 / 36.8 min

Table 9: Adding a module through the seams in Section 3.5.

Property	Measured
Size	11 modules, 789 lines
Edits to the triage decision path	0
How it reads data	duck-typed `all_cases()`; it does not hard-import the store
Imports of the decision core	none (enforced by an AST test, not a convention)
Other changes required	one UI tab; one additive, nullable database column
Stream with an injected outbreak (132 cases)	1 alert, the injected one
Flat stream, no outbreak (120 cases)	0 alerts

pipeline is, we think, the difference between a framework and an application.

5.9 What this would mean in practice

On the sandbox, 42.1% of messages resolve to "care at home", and for urgent cases the naive nearest clinic is unsuitable in 80% of the demonstration villages. The system asks about 2.1 yes/no questions to reach a decision and resolves 74.8% of unclear cases. An explicitly illustrative model, with its assumptions stated in the harness output (one avoided trip is taken as four hours and INR 300), gives per 1,000 callers roughly 421 avoided trips, 1,684 hours and INR 126,316. These are not field measurements and we do not present them as such; they indicate the shape of the benefit that would have to be measured in a deployment, under SDG targets 3.8 and 10.

6 Discussion

Safety lives in the dullest mechanism. The components a reader would expect to carry the result, the conformal gate, the emergency prior boost, the AHP prioritiser, turn out not to. The ablation in Table 3 is unambiguous: remove worst-case labelling over the prediction set and misses go from zero to 6.3%; remove the boost constant and nothing changes. Worst-case labelling is a

one-line rule. It is also the only mechanism that converts the conformal coverage guarantee into a guarantee about the label, which is what the health worker actually sees. We think the general lesson is that in a pipeline like this one, the guarantee should be traced from the property you care about back to the line of code that produces it, and that line is rarely the sophisticated one.

Optimising one component against another's assumptions. The follow-up result is, to us, the most instructive thing in the paper. The question selector was correct. The classifier was correct. Each had been tested in isolation. Together they lost to random selection, because the selector chose questions on the assumption that both answers carry information and the classifier honoured only one of them. Nothing in either module's tests could have caught this; only measuring the composed loop against a dumb baseline did. A policy that cannot beat random is a policy whose assumptions are being violated somewhere, and that is a useful diagnostic to keep.

The same section contains a warning about the obvious fix. Applying absence evidence uniformly across all conditions is textbook Bayes and made the system more alarmed the more a caller denied. Its aggregate metrics were fine. A monotonicity property (a "no" must never widen the set or raise the tier), checked on a single hand-built case, is what caught it. We would recommend that any system which incorporates negative evidence into a presence-only model state and test that property explicitly.

The model is a reader, not a decider. Section 5.1 shows the model making decisions badly and Section 5.5 shows it reading messages well: 96.2% token recall on real code-mixed lay phrasings, and the safety bound intact end to end. The 13.3% "failure rate" that briefly suggested otherwise was our schema bug, not the model's limitation. We take from this that a small local model is a good fit for the part of the problem that needs language understanding, and a poor fit for the part that needs calibrated judgement, and that the architecture should reflect that split rather than ask one component to do both. Safety instructions in a prompt did not produce calibrated behaviour; structure around the model did.

What α is doing. Table 4 is more candid than it is flattering. At the shipped $\alpha = 0.10$ the LAC threshold is so low (0.0216) that a third of cases fall through to "return everything", and the resulting coverage of 1.000 owes as much to that fallback as to conformal filtering. At $\alpha = 0.01$ the gate is a real gate, there is no fallback, and every reported metric is as good or better. We kept 0.10 as the shipped value because it is the conventional one and because the sweep, not the point, is the result: the trade-off between caution and commitment is a designed dial that behaves as designed across its range. A deployment should choose α on its own tolerance for over-calls, and should report the fallback rate alongside coverage whenever this construction is used.

Coordination as the contribution. None of the components is new. What we claim is the arrangement: a three-valued record that refuses to confuse "absent" with "unknown"; a set-valued decision whose size, not whose content, triggers the next action; a question selected deterministically and only phrased by the model; worst-case labelling and a danger-sign override that cut across every branch; and seams that let a second module plug into the same case stream without touching the decision path. The surveillance module in Section 5.8 is the proof of that last point, and the AST test that forbids it from importing the decision core is what keeps the proof true as the code changes.

7 Limitations and future work

We state these rather than leave them to be found.

1. **The dataset is a demonstration dataset, and the labels are ours.** The 41 conditions come from a public teaching tabulation that was not collected from patients. The urgency tiers were assigned by the authors; no clinician reviewed them and they are not anchored to ESI [8], IMNCI, or any published standard. Every rate in this paper is relative to those labels. The work validates the framework's uncertainty-handling mechanism; it does not validate clinical safety. Anchoring the tiers to a published standard with clinician sign-off is the first thing that has to happen before any deployment claim.

2. **Only two emergency conditions exist.** The urgent population ($n = 142$) is dominated by SEVERE, and emergency recall cannot generalise to conditions absent from the dataset.

3. **The sandbox is easier than reality.** Symptom–condition associations are effectively binary (100% hold in at least 70% of a condition's rows; none below). Real associations are probabilistic. The information-gain selector has markedly less headroom here than it would on real data, and the follow-up figures should not be extrapolated. This also explains why the relevance-threshold sweep for the selector is flat: any cut-off in $[0, 0.70]$ selects the same pool.

4. **Conformal coverage is marginal and finite-sample.** There are 304 calibration points. Nominal coverage is slightly under-attained at tight α (Table 4), and at the shipped α the gate falls back to "all candidates" on 33.9% of cases.

5. **Follow-up breaks exchangeability.** Once denials have been folded into the record, the input to the conformal gate is no longer exchangeable with the calibration set, so the α bound after follow-up is approximate rather than exact. That is inherent to adaptively collected evidence. We verify it empirically (post-follow-up misses $0.021 \leq \alpha = 0.10$) rather than assume it, and the absence term is deliberately restricted to the positive-evidence pool (Section 5.4).

6. **The paediatric path is effectively untested by these harnesses.** The IMNCI slice needs exam-only signs (respiratory rate, chest indrawing, skin pinch) that cannot come from a text message and have no interface. It is a small hand-coded set of danger signs, not an IMNCI implementation, and young infants under two months are deliberately not routed to the dataset classifier.

7. **The lay end-to-end subsample is not random.** It is selected by fixture coverage (which symptoms have labelled real phrasings). Its delta against the oracle is meaningful; its absolute rates are not comparable to the full set.

8. **Reply handling is measured over authored forms, not observed caller behaviour.** The routing network is ten synthetic facilities with a mock distance provider. The impact figures are an illustrative model with stated assumptions, not field measurements.

Future work. In rough order of importance: clinician-anchored severity labels and a validation study on real presentations; a real SMS/IVR channel behind the existing `answer_provider` seam (the send half exists; the "block until the next inbound reply" half does not); a degradation controller

for automatic backend failover when the local model is unavailable; real road routing through the existing OpenRouteService adapter; full IMNCI ingestion (ear, malaria and nutrition axes, young-infant ruleset); and growing the surveillance module from a working skeleton into something evaluated against real case streams.

8 Related work

Symptom checkers and language models for triage. Rule-based and probabilistic symptom checkers have been evaluated against clinician judgement for a decade; the audit by Semigran et al. [9] found that triage advice was correct in a little over half of standardised cases and that the tools were consistently risk-averse. Our LLM-only baseline (Section 5.1) reproduces that risk-averse collapse in a different technology, with the difference that the model's over-calling is accompanied by a 16.9% miss rate, so it is not even safely conservative. Our position is not that language models should be kept out of triage but that they should read rather than decide.

Selective prediction and conformal sets. The idea that a classifier should be allowed to abstain goes back to Chow's reject option [4] and has a modern treatment for neural models in Geifman and El-Yaniv [5]. Conformal prediction [2, 3] provides set-valued predictions with distribution-free marginal coverage; the least-ambiguous-set construction we use is from Sadinle, Lei and Wasserman [1]. What we add is not a new construction but a specific way of acting on set size: a singleton commits, a small set escalates to its worst member and asks, a large set abstains. The guarantee in Section 5.3 follows directly from coverage once the label is the worst case over the set, and we have not seen that step stated as a design rule for triage.

Prioritisation. The Analytic Hierarchy Process [6] is a standard multi-criteria method; we use it only to order cases within a tier, which is the role the Emergency Severity Index [8] assigns to its acuity levels, and we derive its weights from the comparison matrix rather than transcribe them.

Structuring language-model pipelines. Our extraction stage uses a small open model [10] with multilingual sentence embeddings [11] and an approximate nearest-neighbour index [12] for vocabulary matching. The broader pattern, a language model confined to a well-typed boundary with deterministic logic on either side, is common engineering practice; what we tried to contribute is a worked account of where, in one such pipeline, the dependability actually comes from, with the failures that measurement exposed along the way.

9 Conclusion

A small language model can read a code-mixed symptom message well and cannot be trusted to decide what to do about it. The framework described here keeps the model in the first role and surrounds it with deterministic structure for the second: a three-valued record, a set-valued classifier whose set size drives the next action, worst-case labelling that turns coverage into a bound on missed urgent cases, and a follow-up loop that asks one question when the set is too large and decides again. On a demonstration dataset the full pipeline misses no urgent case, a property that holds by construction and survives real extraction from lay language, and the follow-up loop then recovers most of the over-caution that property costs.

The results we would most want another team to take from this are the ones we did not expect. The elaborate mechanisms were not where safety lived. A correct question selector and a correct classifier together lost to random selection. The textbook way to use negative evidence made the system more alarmed the more a caller said no. And the language model's apparent failure rate was a bug in our own code. Each of those was found by measuring the composed system against a baseline it should have beaten, and not by reasoning about the components.

Nothing here is a clinical result. The labels are ours, the dataset is synthetic, and the next step is a clinician. The framework, the harnesses, and the provenance of every number are public so that step can be taken by someone other than us.

Reproducibility

Code, data provenance, and every evaluation harness are at `https://github.com/Rishii-10/GMU`. Each number in this paper is read from a tracked results file regenerated by one named script; the deterministic harnesses reproduce byte for byte, and the three that call the model are reproducible to the model's measured determinism of 1.000 at temperature zero.

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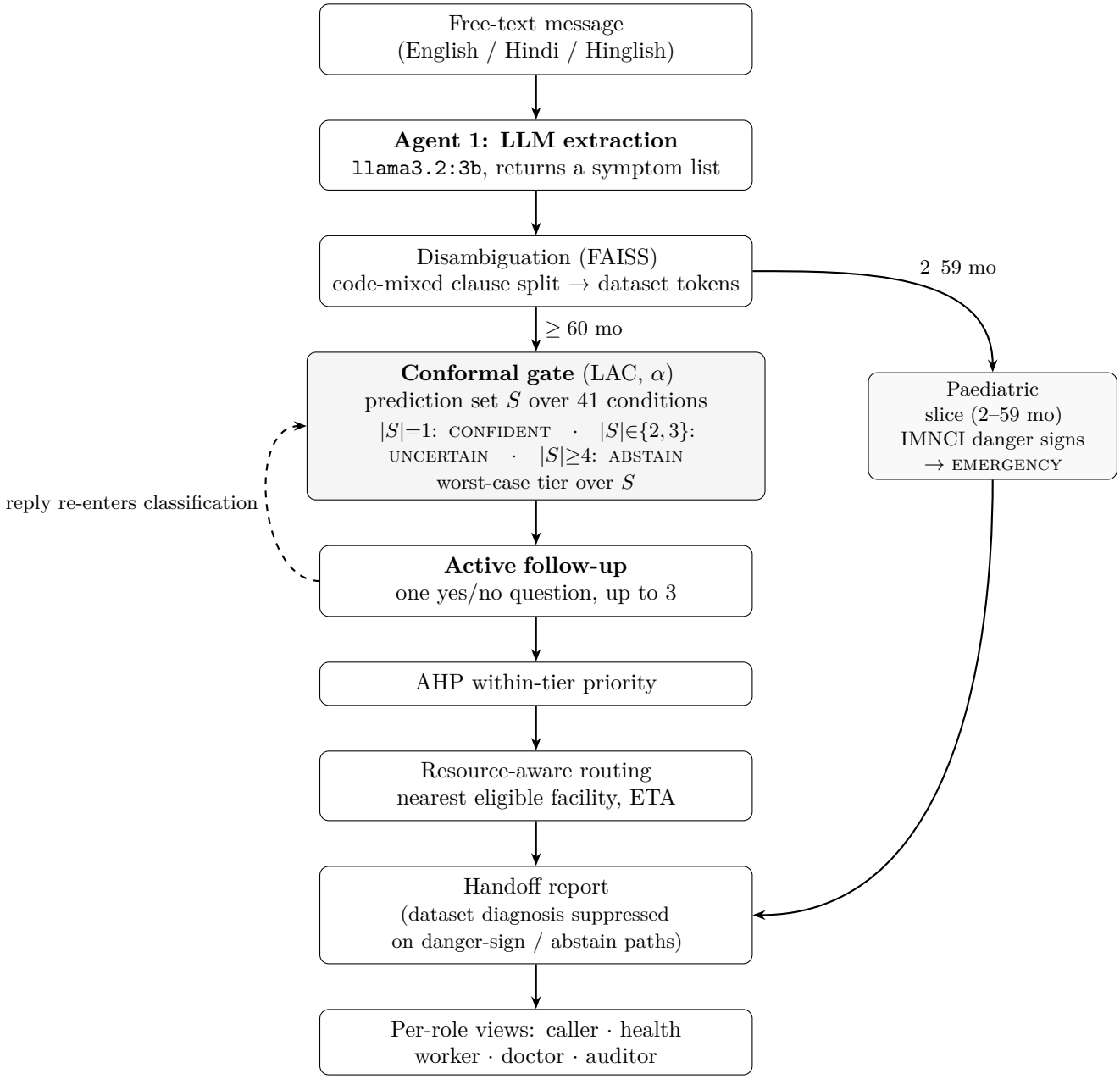


Figure 1: The pipeline. Solid arrows are the forward path; the dashed arrow is the coordination the paper is about: a follow-up answer re-enters classification rather than being bolted onto the output. Worst-case labelling on the UNCERTAIN and ABSTAIN branches, and the danger-sign override on every branch, are what carry the safety property.

A Supplementary diagrams (new — not part of the original paper)

These figures are *additive*: the paper above is unchanged, and the diagrams are collected here so that what is new is easy to audit. Together they retell the system the way it was built, as a software-engineering exercise rather than a list of components — state the problem and decompose it, show the naive approach and where it breaks, then address each failure one layer at a time. They are cross-referenced to the sections that contain the corresponding prose and numbers.

A.1 The problem, decomposed into four computational tasks

We begin where a build begins: with the problem and its decomposition. Message-based triage is not one task but four that have to hold together (Section 2). Figure A1 states them as a context diagram. The language model is confined to a single task (T1); the one arrow that makes this a *framework* rather than a straight pipeline is the follow-up loop on T2.

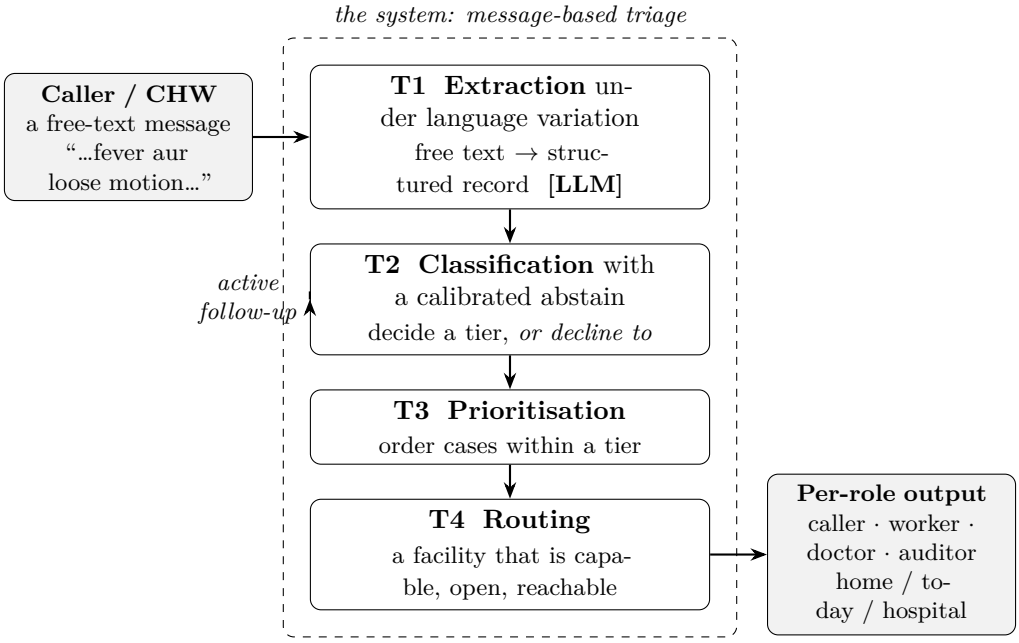


Figure A1: The problem, decomposed into four computational tasks that must be solved together. Only T1 calls the language model; the dashed self-loop on T2 is active follow-up, the mechanism that turns "the classifier is unsure" into "ask one question and decide again". This is the problem statement the rest of the framework answers (Section 2).

A.2 The naive approach, and why it fails

The obvious first implementation is to let the model decide. Figure A2 shows it and the measured failure that motivates everything else (Section 5.1): asked to pick a tier, the model collapses onto one answer and is unsafe and uninformative at the same time.

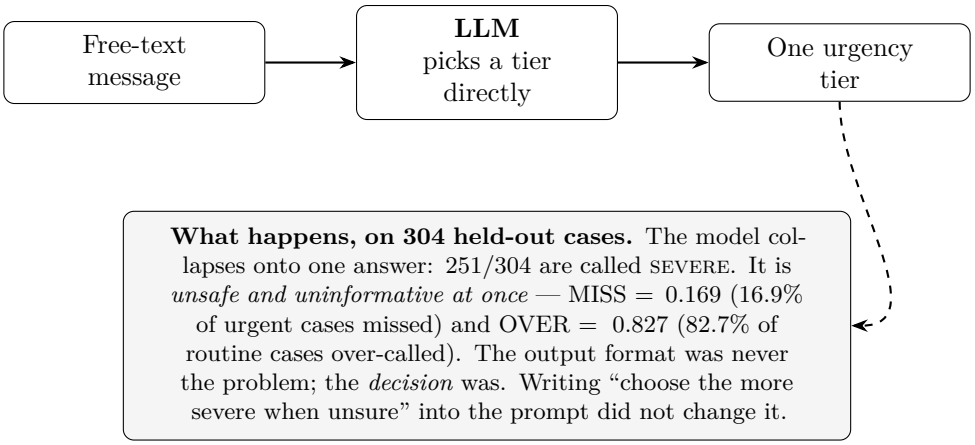


Figure A2: The naive approach, and why it fails. Asking the language model to *decide* produces a risk-averse collapse that is both unsafe and uninformative. This is the motivation for surrounding the model with deterministic structure instead of trusting its judgement (Section 5.1).